Course One
Foundations of Data Science

Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- [x] ~~Complete the questions in the Course 1 PACE strategy document~~
- [x] ~~Answer the questions in the Jupyter notebook project file~~
- [x] ~~Complete coding prep work on project's Jupyter notebook~~
- [x] ~~Summarize the column Dtypes~~
- [x] ~~Communicate important findings in the form of an executive summary~~

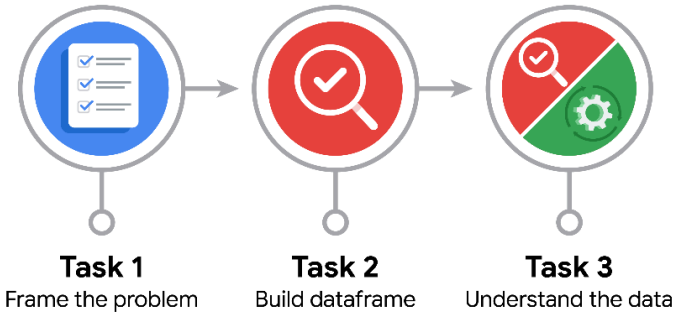
Relevant Interview Questions

Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?

Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- How can you best prepare to understand and organize the provided information?

Review the **Data Dictionary** to define each variable and its constraints. Identify the **primary objective**: building a model to distinguish "claims" from "opinions" to assist moderators. Map out the **stakeholders**: technical (Willow, Orion) vs. non-technical (Mary Joanna, Maika) to tailor future communication.

- What follow-along and self-review codebooks will help you perform this work?

Utilize **pandas documentation** for DataFrame manipulation (specifically `groupby` and `agg`). Refer to the **Google Data Analytics Professional Certificate** materials on the PACE framework for workflow consistency.

- What are some additional activities a resourceful learner would perform before starting to code?

Review **internal meeting notes** to understand the specific concerns of leadership, such as the need for statistical testing later. Search for **TikTok's Trust & Safety** public guidelines to understand the context of "claims" vs. "opinions" in real-world moderation

PACE: Analyze Stage

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

The current data (video duration, view counts, transcription text) provides a solid foundation for initial classification. However, the **298 null values** in key columns like `claim_status` must be addressed or removed before the data is truly "sufficient"

- How would you build summary dataframe statistics and assess the min and max range of the data?

Use `.info()` to check for non-null counts and data types (Dtypes). Apply `.describe()` to view the `min`, `max`, `mean`, and `std` for all numerical engagement columns

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

The **mean** view count (~254,708) is much higher than the **median** (~9,954), indicating highly skewed data. Most engagement metrics (views, likes, shares) show a massive interval between the 75th percentile and the maximum value, suggesting viral outliers

PACE: Construct Stage

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.

PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploratory data analysis?

Investigate why **298 rows** are missing data. Is it a technical glitch in data collection, or are these specific types of videos (e.g., deleted content)? Confirm if the `video_transcription_text` is raw or pre-processed, as this will impact NLP model construction.

- What data initially presents as containing anomalies?

Videos with **hundreds of thousands of likes** but very few views (or vice-versa) should be flagged as potential bot activity or data errors. The high engagement levels associated with **banned authors** suggest a strong correlation between controversial content and virality

- What additional types of data could strengthen this dataset?

User Report Reason: Knowing *why* a user flagged a video (e.g., "misinformation" vs. "harassment") would help refine the claim/opinion distinction. **Timestamp Data:** Knowing when a video was posted could help identify if claims are more common during specific global events.